NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 02

Technical Requirements

Architecture, technology stack, interfaces, quality attributes and operational requirements.

Document control

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Evidence labels used in this document: Implemented verified in code, configuration or test output; Inferred reasoned from evidence but not executed; Recommendation proposed, not implemented; Open question needs a decision; Unknown no evidence available; Estimate planning figure, not a measurement.

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1 System context

Version 2.0 update. Part 3 added Google SSO (Credential Manager plus `google-auth-library`), FCM push (`firebase-messaging` plus `firebase-admin`), WorkManager reminders, full en/zu/tn resources, release signing, Zod 4 and Node 22. The diagram shows the core components; the new pieces are listed in the tables below.

Nexus has two deployable parts: the Android app (`app/`, package `com.example.nexus`) and the Nexus API (`backend/server.js`). The app talks to the Nexus API over REST/JSON. Since v2.0 it also uses Google Identity (Credential Manager) for SSO and Firebase Cloud Messaging for push. A Retrofit client for the public MyMemory translation API is defined, but nothing in the app calls it (Implemented, AUD-09).

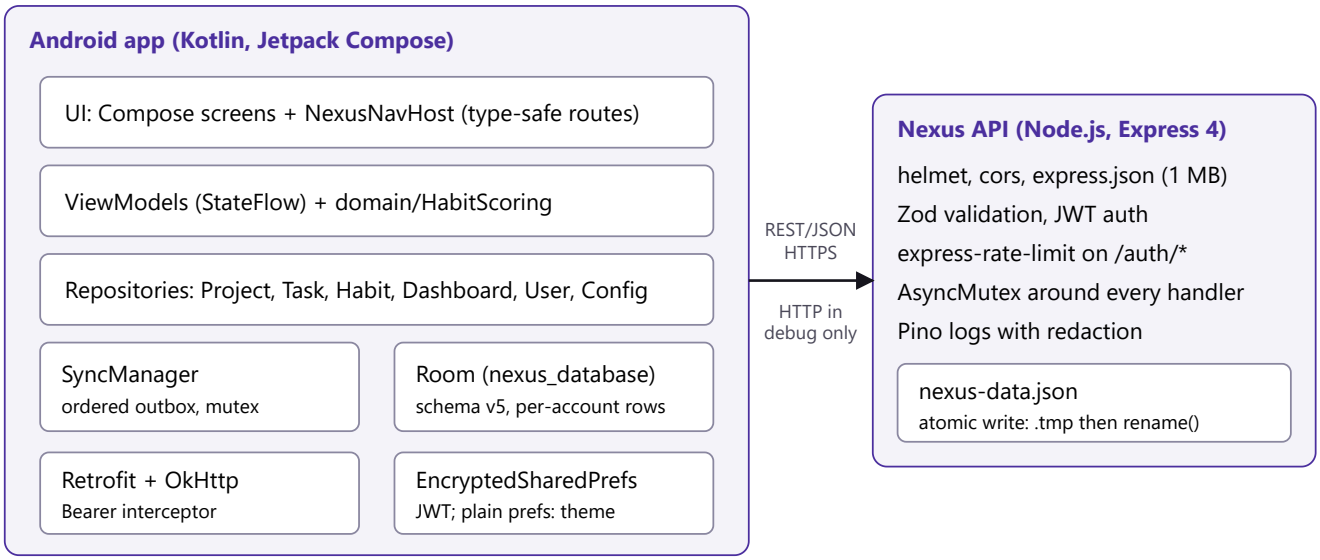


Figure 1. Component view at the evidence baseline. Arrows show request direction only.

2 Technology stack

Versions as declared in build files

Layer	Technology and version	Source
Android language / build	Kotlin 2.0.21, AGP 9.3.1, Gradle 9.7.1 wrapper, JDK 17 target, KSP 2.0.21-1.0.28	`build.gradle.kts`, `gradle-wrapper.properties`
SDK levels	minSdk 24, targetSdk 37, compileSdk 37; versionCode 1, versionName 1.0	`app/build.gradle.kts`
UI	Jetpack Compose (BOM 2026.02.01), Material 3, Navigation Compose 2.8.0 with kotlinx-serialization routes, material-icons-extended	`libs.versions.toml`, `app/build.gradle.kts`
Networking	Retrofit 3.0.0 + Gson converter, OkHttp logging-interceptor 4.12.0 (BASIC in debug, NONE in release)	`RetrofitClient.kt`

Layer	Technology and version	Source
Local storage	Room 2.8.5 (runtime, ktx, compiler via KSP); AndroidX Security Crypto 1.1.0	`Database.kt`, `TokenManager.kt`
SSO and notifications (v2.0)	androidx.credentials 1.6.0 + play-services-auth, googleid 1.1.1; WorkManager 2.12.0; Firebase BoM 35.1.0 (firebase-messaging); google-services plugin 4.5.0, applied only when `app/google-services.json` exists	`app/build.gradle.kts`
Quality tooling	ktlint plugin 12.1.2, detekt 1.23.8, Kover 0.9.1; JUnit 4.13.2, MockK 1.13.12, coroutines-test 1.8.1, Compose UI test	`build.gradle.kts`
API runtime	Node.js 22 (CI and Docker; required by google-auth-library 11 and firebase-admin 14), ES modules	`.github/workflows/*.yml`, `backend/Dockerfile`
API libraries	express ^4.21.2, bcryptjs ^2.4.3, jsonwebtoken ^9.0.2, zod ^4 (4.6.7 installed), google-auth-library ^11.3.3, firebase-admin ^14.5.0, express-rate-limit ^7.5.0, helmet ^8.3.0, cors ^2.8.6, pino ^10.3.1, pino-http ^11.0.0	`backend/package.json`
API tests	Jest ^29.7.0 and Supertest ^7.0.0, run in band with `--experimental-vm-modules`	`backend/package.json`

3 Architecture requirements

Technical requirements

ID	Requirement	How it is met
TR-01	Single-activity MVVM with feature packages.	`MainActivity` hosts `NexusNavHost`. UI is grouped under `ui/<feature>`; ViewModels expose `StateFlow` UI state **Implemented**.
TR-02	Repositories own all data access; ViewModels never call Retrofit directly.	Six repositories are created once in `NexusApp` (manual DI container). ViewModels take constructor parameters with defaults so tests can inject fakes **Implemented**.
TR-03	Business rules live outside the UI.	`domain/HabitScoring` is pure Kotlin and is unit-tested **Implemented**.
TR-04	Local writes are transactional with their outbox record.	`SyncManager.save()` runs the entity write and the `PendingChange` insert inside `withTransaction`, under a coroutine `Mutex` **Implemented**.
TR-05	Reads never overwrite unsent local edits.	`SyncManager.read()` flushes the queue first and falls back to Room if the flush fails or the network is unavailable **Implemented**.
TR-06	Room schema changes ship with migrations; no destructive fallback.	Migrations 3→4 and 4→5 are registered; `fallbackToDestructiveMigration` is deliberately absent **Implemented**. Schema export is off (AUD-12).
TR-07	API base URL is configured at build time.	`NEXUS_API_URL` Gradle property or environment variable must match `https?://host[:port]/path/`. It defaults to `http://10.0.2.2:5263/api/`, and release builds fail unless it starts with `https://` **Implemented**.
TR-08	API validates every write before it touches the store.	Zod schemas for register, login, profile (strict), password, project, task, habit and sync. Failures return 400 `VALIDATION_ERROR` **Implemented**.

ID	Requirement	How it is met
TR-09	API errors use one response shape.	Central handler returns `{ error: { message, requestId, code? } }`; in production a 500 message is replaced with "Internal Server Error" **Implemented**.
TR-10	Idempotent offline replay.	`POST /api/sync` takes `operationId` (UUID) and stores receipts per user. A repeated operation returns 204 without effect **Implemented**.

4 Interfaces

4.1 Nexus API endpoints

All paths are under `/api` except the root `/health`. "Auth" means a `Authorization: Bearer <JWT>` header is required; a missing or invalid token gives 401 `UNAUTHORIZED`, and a token for a deleted user gives 401 `SESSION_EXPIRED`.

Routes defined in backend/server.js

Method and path	Auth	Purpose and notable responses
`GET /health`, `GET /api/health`	No	`{status:"ok", uptime}`
`GET /app/config`	No	Static config: versions 1.0.0, maintenance off, feature flags, no announcements
`GET /localization/:language`	No	Five-string bundle for en/zu/tn (falls back to en)
`POST /auth/register`	No, rate-limited	201 `{token}`; 409 duplicate email; 429 after 10 per 15 min
`POST /auth/login`	No, rate-limited	200 `{token}`; 401 invalid credentials (emails case-insensitive)
`POST /auth/google`	No, rate-limited	`{idToken}`; 201 new account or 200 existing/linked, `{token}`; 401 `INVALID_GOOGLE_TOKEN`; 503 `SSO_NOT_CONFIGURED`
`POST /users/me/devices`, `DELETE /users/me/devices/:token`	Yes	Register or unregister an FCM token (204)
`POST /users/me/notifications/test`	Yes	Pushes a test notification; returns `{configured, sent, devices}`
`GET` / `PATCH /users/me`	Yes	Profile `{email, displayName, language, notificationsEnabled}`; PATCH rejects unknown fields
`POST /users/me/password`	Yes	204; 401 wrong current password
`DELETE /users/me`	Yes	204; removes the user and all their data
`GET /dashboard`	Yes	`{projects, tasks, activity}` counts
`POST /sync`	Yes	204 applied or duplicate; 400 invalid payload; 409 `SYNC_CONFLICT`
`GET` / `POST /projects`	Yes	List / create (201)
`GET` / `PUT` / `DELETE /projects/:projectId`	Yes	Read / update / delete (delete cascades to tasks); 404 `PROJECT_NOT_FOUND`

Method and path	Auth	Purpose and notable responses
`GET` / `POST /projects/:projectId/tasks`	Yes	List / create task; 404 if the project is not owned
`PUT` / `DELETE /tasks/:taskId`	Yes	Update / delete; 404 `TASK_NOT_FOUND`
`GET` / `POST /habits`, `PATCH` / `DELETE /habits/:habitId`	Yes	List / create / update / delete; 404 `HABIT_NOT_FOUND`

When `NODE_ENV=test` only, `GET /api/test/error-500` also exists for error-handler tests **Implemented**. The app uses the direct project, task and habit endpoints only when no `SyncManager` is wired in. In the production wiring (`NexusApp`), writes go through `POST /sync` **Implemented**.

4.2 Device integrations

- Permissions: `INTERNET`, `ACCESS_NETWORK_STATE` and (v2.0) `POST_NOTIFICATIONS`, requested at runtime on Android 13+ **Implemented**.
- WorkManager periodic job (reminders) and an FCM `FirebaseMessagingService` (v2.0) **Implemented**.
- `ConnectivityManager` network callback triggers sync when a network becomes available **Implemented**.
- Android share sheet for CSV export (`Intent.ACTION_SEND`, `text/csv`) **Implemented**.
- Backup: `allowBackup="true"`; since v2.0 both shared preferences and the Room database are excluded from cloud backup and device transfer (AUD-05 fixed).

5 Quality attributes

Quality attributes and current position

Attribute	Current position	Related
Security	Controls listed in NFR-01 and NFR-02 are implemented. CORS is open (`*`), which is documented as intentional for a mobile client. No formal security review or penetration test has been performed **Unknown**.	AUD-06, AUD-16
Reliability	Atomic file writes and an in-process mutex. However, a JSON file that cannot be parsed at startup is replaced on the next write (AUD-01).	NFR-04
Scalability	Single process and single file; every request, including bcrypt hashing, holds one global lock. `PROJECT_IMPROVEMENTS.md` treats this as adequate for fewer than 50 users and 10,000 tasks **Inferred**; this was not load-tested.	CON-01, AUD-07
Maintainability	Feature packages, shared test fakes, ktlint/detekt/Kover configured. Lint is not enforced in CI (AUD-04).	NX-07-CICD
Portability	Android 7.0+. The API runs on any Node host or as a container.	NX-11-DEP

6 Operational requirements

API runtime configuration (names only; values are never stored in the repository)

Variable	Required	Behaviour
`JWT_SECRET`	Yes	At least 32 characters and not a known placeholder, or the process exits with code 1. Keep it stable across restarts, or existing sessions become invalid.

Variable	Required	Behaviour
`PORT`	No (5263)	HTTP listen port.
`DATABASE_PATH`	No	Data file path; defaults to `backend/nexus-data.json`, and to `/data/nexus-data.json` in the Docker image.
`NODE_ENV`	No	`production` hides 500 messages; `test` disables listening and enables the test route.
`LOG_LEVEL`	No (info)	Pino log level.
`GOOGLE_CLIENT_ID`	For SSO	Web OAuth client ID(s), comma-separated, accepted as the ID-token audience. Unset disables `/auth/google` (503).
`FIREBASE_SERVICE_ACCOUNT_BASE64`	For push	Base64 of the Firebase service-account JSON (or use `GOOGLE_APPLICATION_CREDENTIALS`). Unset makes push a logged no-op.
`TRUST_PROXY`	Behind a proxy	Express trust-proxy setting (e.g. `1`) so rate limiting uses the client IP.

- **OPS-01** TLS termination must be provided by the host or a reverse proxy; the API itself serves plain HTTP Implemented.
- **OPS-02** The data file (or `/data` volume) must be persistent and backed up. The repository contains no backup tooling Implemented.
- **OPS-03** Run exactly one API instance per data file (CON-01).
